



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 10 • STANDARD 6.G.4**

# Surface Area of Pyramids

Lesson 10-5 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

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## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



*Like the Great Pyramid of Giza — a square on the bottom, 4 triangles slanting up to a point*

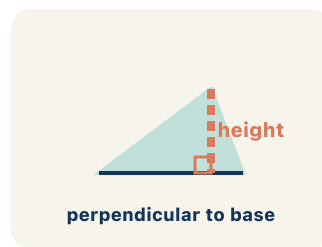
### Pyramid

Write the definition:

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*If you slide your finger from the bottom edge up the triangle face to the top point — that distance is the slant height*

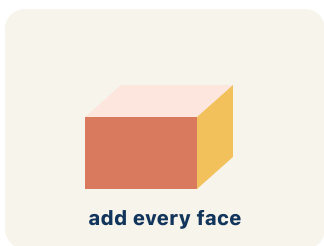
### Slant height

Write the definition:

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*A square pyramid has 4 lateral faces — one triangle for each side of the square base*

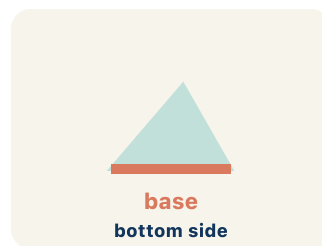
### Lateral face

Write the definition:

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*A square pyramid sits on a square base; base area = side × side*

### Base

Write the definition:

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*The pointy top of a pyramid — all the slanted edges connect here*

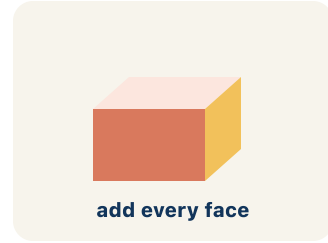
### **Apex**

**Write the definition:**

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*For a square pyramid: lateral area =  $4 \times (\frac{1}{2} \times \text{base edge} \times \text{slant height})$*

### **Lateral area**

**Write the definition:**

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can find the surface area of a pyramid by adding the base area and the lateral faces.**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area



Tap any word to see what it means and a picture.

1

A solid with a polygon base and triangular faces meeting at one point is a

2

The height of a triangular face of a pyramid, measured along the slanted side, is the

3

A triangular side face of a pyramid is a .

4

The polygon on the bottom of a pyramid is the .

5

The single top point where the triangular faces of a pyramid meet is the

6

The combined area of all the side faces, not counting the base, is the



### Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

- A.  $95 \text{ in}^2$
- B.  $70 \text{ in}^2$
- C.  $95 \text{ in}^3$
- D.  $120 \text{ in}^2$

**Step 1** Base:  $5 \times 5 = 25 \text{ in}^2$ .

**Step 2** Each lateral face:  $\frac{1}{2} \times 5 \times 7 = 17.5 \text{ in}^2$ .

**Step 3** Four faces:  $4 \times 17.5 = 70 \text{ in}^2$ .

**Step 4**  $SA = 25 + 70 = 95 \text{ in}^2$ .

 **Answer:** A.  $95 \text{ in}^2$

 **Try — put the steps in order**

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

$SA = 25 + 70 = 95 \text{ in}^2$ .

Base:  $5 \times 5 = 25 \text{ in}^2$ .

Four faces:  $4 \times 17.5 = 70 \text{ in}^2$ .

Each lateral face:  $\frac{1}{2} \times 5 \times 7 = 17.5 \text{ in}^2$ .

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** How many lateral (triangular) faces does a square pyramid have?

- A. 4
- B. 3
- C. 5
- D. 6

**Step 1**

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**Step 2**

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**Step 3**

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**Answer:**

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 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** What is the area of one triangular face with base 8 in and slant height 6 in?

- A.  $24 \text{ in}^2$
- B.  $48 \text{ in}^2$
- C.  $14 \text{ in}^2$
- D.  $24 \text{ in}^3$

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?**

- A.  $20 \text{ in}^2$
- B.  $40 \text{ in}^2$
- C.  $13 \text{ in}^2$
- D.  $20 \text{ in}^3$

Show your work:

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**2. A gift box is a square pyramid with a base edge of 9 in. What is the area of its square base (the part that sits on the table)?**

- A.  $81 \text{ in}^2$
- B.  $36 \text{ in}^2$
- C.  $18 \text{ in}^2$
- D.  $324 \text{ in}^2$

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Two pyramids have the same base area ( $64 \text{ in}^2$ ) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?**

*Sentence starter: Pyramid A:  $SA = 64 + 4(\frac{1}{2} \times 8 \times 6) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$ . Pyramid B:  $SA = 64 + 4(\frac{1}{2} \times 8 \times 10) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$ . Pyramid B has  $\underline{\hspace{1cm}}$  more  $\text{in}^2$  because  $\underline{\hspace{1cm}}$ . Slant height only affects  $\underline{\hspace{1cm}}$  because  $\underline{\hspace{1cm}}$ .*

Show your work:

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## Reflect — Exit Ticket

**A square pyramid has a base edge of 6 in and a slant height of 5 in. What is the total surface area?**

- A.  $96 \text{ in}^2$
- B.  $60 \text{ in}^2$
- C.  $96 \text{ in}^3$
- D.  $36 \text{ in}^2$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Pyramid
2. **Notes 2:** Slant height
3. **Notes 3:** Lateral face
4. **Notes 4:** Base
5. **Notes 5:** Apex
6. **Notes 6:** Lateral area
7. **We Try (notes):** A. 4 — *A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.*
8. **You Try (notes):** A.  $24 \text{ in}^2$  — *Area of a triangle =  $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 6 = 24 \text{ in}^2$ .*
9. **Try It 1:** A.  $20 \text{ in}^2$  — *Area of a triangle =  $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 5 = 20 \text{ in}^2$ . You use the slant height (5) as the triangle's height, not the pyramid's vertical height.*
10. **Try It 2:** A.  $81 \text{ in}^2$  — *The base is a square, so base area = side  $\times$  side =  $9 \times 9 = 81 \text{ in}^2$ . You would still add the four triangular faces to get the full surface area.*
11. **Exit Ticket:** A.  $96 \text{ in}^2$  — *Base:  $6 \times 6 = 36 \text{ in}^2$ . Each lateral face:  $\frac{1}{2} \times 6 \times 5 = 15 \text{ in}^2$ . Four faces:  $4 \times 15 = 60 \text{ in}^2$ . SA =  $36 + 60 = 96 \text{ in}^2$ . Surface area uses square units ( $\text{in}^2$ ).*